Universidade de Brasília
Faculdade do Gama

Digital Integrated Circuit Design

Laboratory 3 – CMOS Logic and Hierarchical Design

4:1 Multiplexer

Choose one of the topologies shown in **Figure 1** and design and simulate a 4-input multiplexer using two **2:1 inverter multiplexers**, as shown in **Figure 2**.

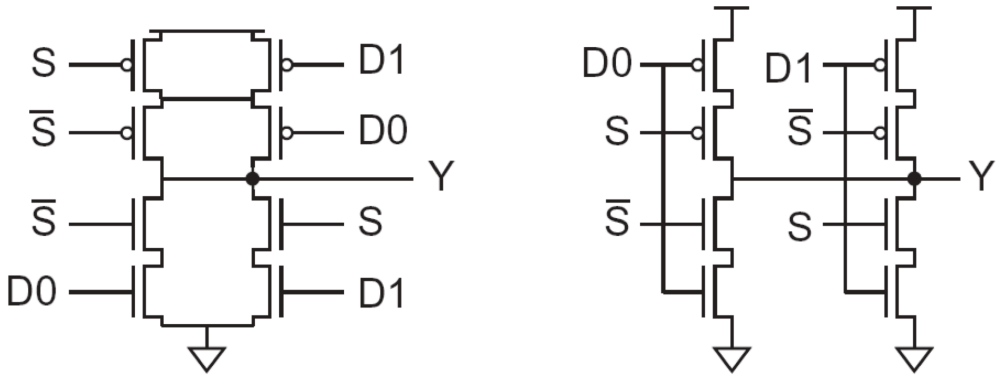


Figure 1

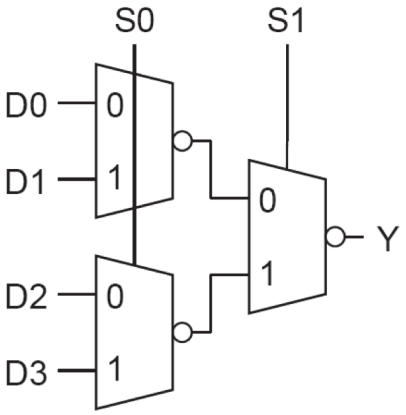


Figure 2

4-bit Ripple Carry Adder

Design and simulate a **4-bit ripple carry adder** using 4 **full adders** (**Figure 3**).

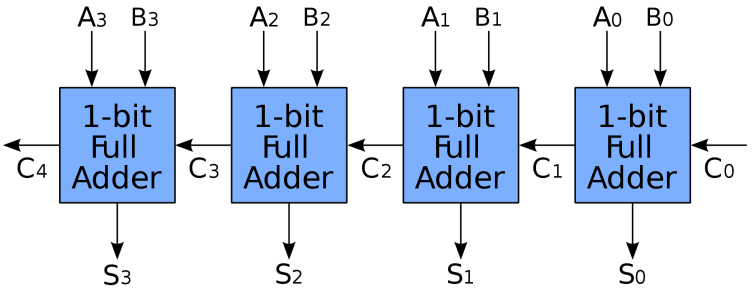


Figure 3

Each **full adder** must be implemented according to the schematic shown in **Figure 4.**

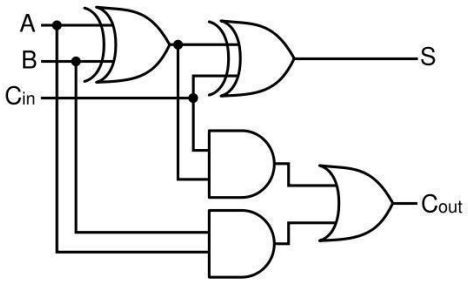


Figure 4

The **AND2** and **OR2** gates can be derived from **NAND2**, **NOR2**, and **INVERTER** gates. The **XOR** gate can be designed according to the schematic in **Figure 5**.

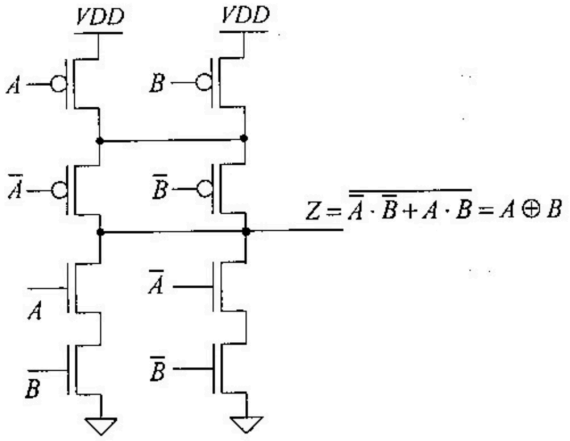


Figure 5

Task

Write a short report showing the schematics and simulations of the following cells:

- mux2;
- mux4;
- XOR2;

- full adder;
- 4-bit ripple carry adder.

Describe the results for each cell.